Exercise and Chronic Kidney Disease

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A consistent exercise routine helps people with chronic kidney disease feel better and improve medical conditions like diabetes and heart disease.

Connection between exercise and kidney health

Exercise is very helpful when you have kidney disease because it helps to improve the health of your kidneys and other organs like your heart. Making exercise part of your daily or weekly routine will help you build muscle and strength, increase your energy levels, and your ability to do the daily activities needed to care for yourself, your family, and your home.

Exercise has the following benefits for people with kidney disease:

- Improved heart function
- Better blood pressure management
- Reduced risk for developing diabetes
- Improved blood sugar levels if you have diabetes
- Improved muscle growth and strength
- Decreased anxiety and depression
- Improved weight management

Some research has shown that exercise may help improve kidney function and anemia.

Aerobic exercise, also known as cardio, increases your heart rate and breathing rate. It involves rhythmic, repetitive movements like running that use large muscle groups. It improves heart and lung function and the health of blood vessels, all of which are important for preserving kidney health.

Keep in mind that your heart is a muscle that needs exercise like other muscles, and your lungs depend on many muscles for you to breathe properly.

Weight bearing exercises using weights or resistance bands help build and maintain your bones and muscles. Preserving bone health is important at all stages of kidney disease, and especially for people taking glucocorticoids like prednisone, which can lead to osteoporosis. Having more muscle makes it easier to maintain a healthy weight because it burns more calories at rest than fat. Maintaining a healthy weight is better for kidney health.

Some basics about exercise for people with chronic kidney disease

The saying "use it or lose it" is true, especially for people with kidney disease who are at increased risk of losing muscle mass. Staying in bed or sitting around all the time leads to less muscle strength and muscle mass, making it harder to do basic tasks. Over time, you may end up not being able to move on your own and you could develop disability. If you get moving, you can maintain and even increase your strength and keep active.

Regular exercise is important for people at all stages of kidney disease, including those on dialysis, and those who have had a transplant. Exercise should be part of your treatment plan at every point in your journey with kidney disease, including while you're waiting for a transplant. Building muscle mass and improving muscle function, including that of lung and heart muscle, is crucial for a good outcome from surgery.

You should plan to do some type of exercise on most, if not all, days of the week. Exercise includes physical activities that engage your large muscles like gardening and walking. Discuss with your doctor and healthcare team about what types of exercise you can safely do. Think about the types of exercise that you could enjoy the most and fit into your schedule so that you'll have an easier time of being consistent.

People on dialysis who exercise regularly report that they sleep better, have more energy and muscle strength, and are better able to do daily tasks and activities.

Exercising using foot bikes and other devices during hemodialysis is available in some dialysis units. Ask your

dialysis provider about these options.

Healthcare and exercise professionals that can help with your exercise goals

- Physical therapists/occupational therapists: licensed healthcare professionals who create personalized exercise programs to improve strength, range of motion, and endurance. Usually work in a healthcare setting such as a hospital or outpatient clinic.
- Exercise physiologists: certified healthcare professionals who create personalized exercise programs for people with chronic diseases or health conditions. Usually work in a healthcare setting like a hospital or outpatient clinic.
- Cardiac rehab specialists: a certified healthcare professional who makes a personal plan for recovery after a heart attack. Usually work in a healthcare setting like a hospital or outpatient clinic.
- Athletic trainers: helps to improve functional outcomes and specializes in patient education to prevent injury and re-injury. Usually work in school-based sports programs or sports teams. May have a health-related degree or certification.
- Personal trainer: creates customized exercise workout plans, usually in the setting of a gym. May have a certification.

Points to keep in mind

- Talk to your doctor and healthcare team before starting an exercise program
- Find activities that you find enjoyable so that you can keep consistent
- Eat and drink according to recommendations from your doctor and healthcare team. This is especially important if you have diabetes and are at risk of low blood sugar.
- Stay safe – have the right gear, start slow, warm up, breathe.

Questions for your healthcare team

- Do you know where I can find free or low-cost exercise classes or gym memberships?
- What types of exercise classes are appropriate for me and which exercise professionals should I work with to achieve my health goals?

- [If you are on dialysis] What types of exercise are safe for someone with my type of dialysis access?
- How much fluid should I have before, during, and after I exercise?
- What food should I eat, and types of fluids should I drink before, during, or after exercise?

More Resources

- Chronic Kidney Disease, Dialysis and Keeping Fit: A Practical Guide
- Staying Fit with Kidney Disease
- NKF Nutrition Hub

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